



12.6% Fewer Vehicles, Zero Regressions: An Audit-First ALNS for Electric-Vehicle Routing

Reproducing & extending Schneider, Stenger & Goeke (2014) E-VRPTW — constraint-guided ALNS, exact charging subproblems, and stage-gate audits with independent evidence replay

Scan for code & evidence

github.com/Oneblaze/FURP-2026-Yiyang-GUO-EVRP-TW



–12.6%

fleet vehicles

best vehicles 87 → 76
over 12 instances

–5.1%

total distance

7,744 → 7,349 units

36 / 36

feasible formal runs

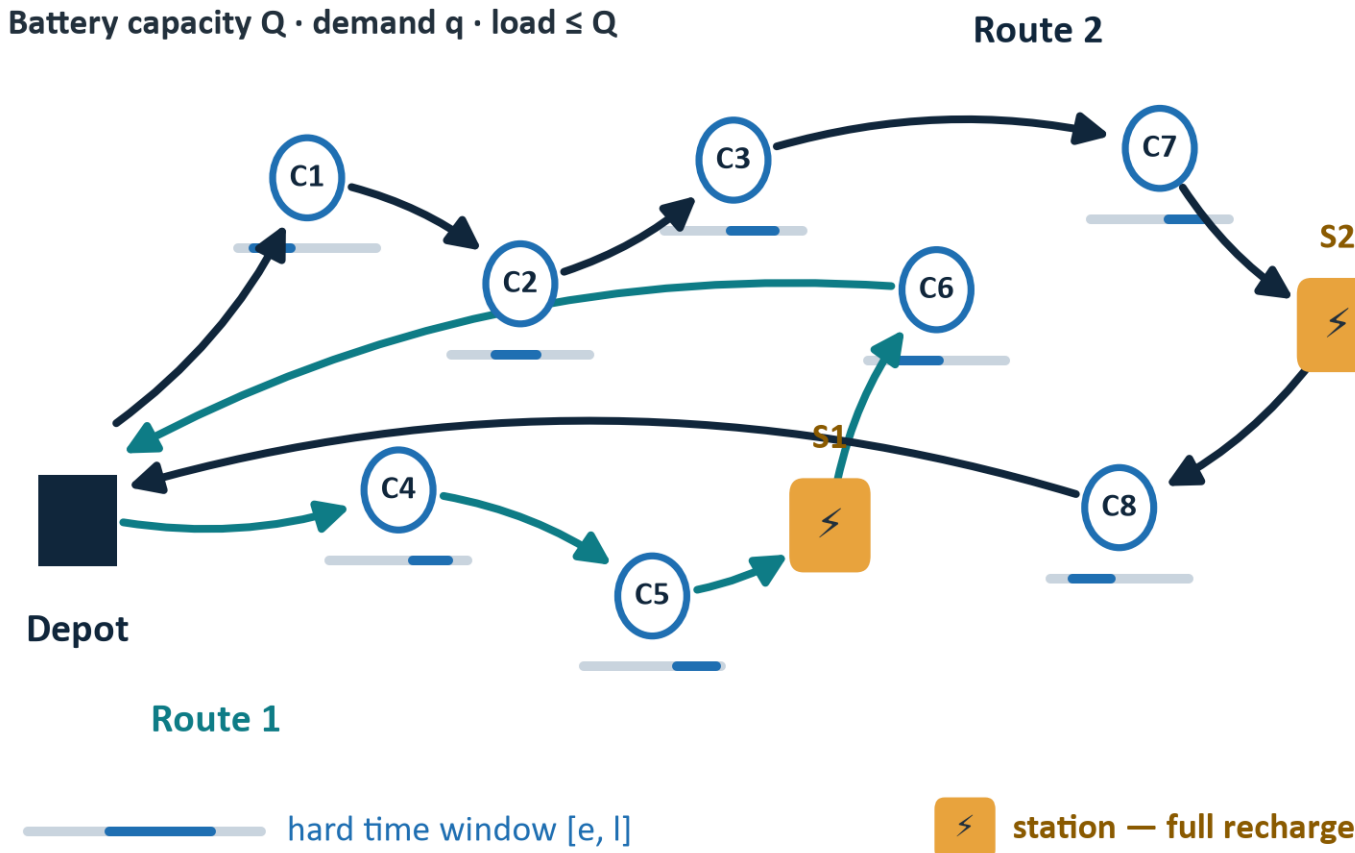
zero objective regressions

–22.0%

exact-charging CPU

median saving · C instances

The Problem — E-VRPTW



8 customers · 2 stations · hard time windows — feasibility hinges on exact charging decisions

Method — ALNS Engine



Protocol — Stage-Gated & Audited

every stage gated by an independent CLI review that replays raw JSON + event logs

- Stage 0**
Baseline reproduction of SSG (2014) — 36/36 feasible, frozen forever
- Stage 1**
Lexicographic objective: vehicles $\gg$ distance $\gg$ charging time
- Stage 2.1–2.3**
Route reduction → constraint-guided ALNS
–12.6% vehicles · –5.1% distance · zero regressions
- Stage 3.0–3.4**
Cheap screening · cache · exact deadline · control parallel
–22% median exact-charging CPU (C set)
- Stage 4**
Adaptive operator weights — wins on 7/36 runs (144/144 re-verified)
- Stage 5.1**
Honest BKS audit — 92/124 entries model-incompatible, no fake gaps

READY

READY

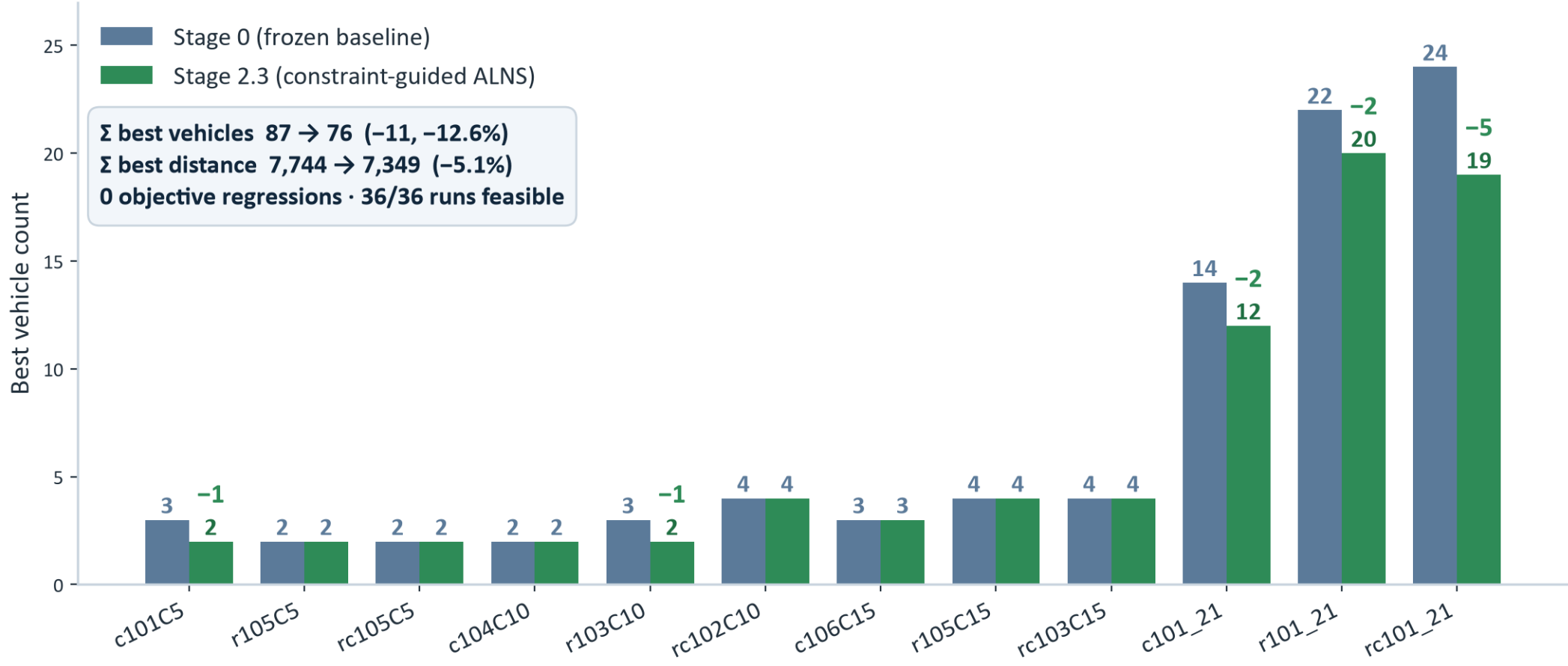
READY

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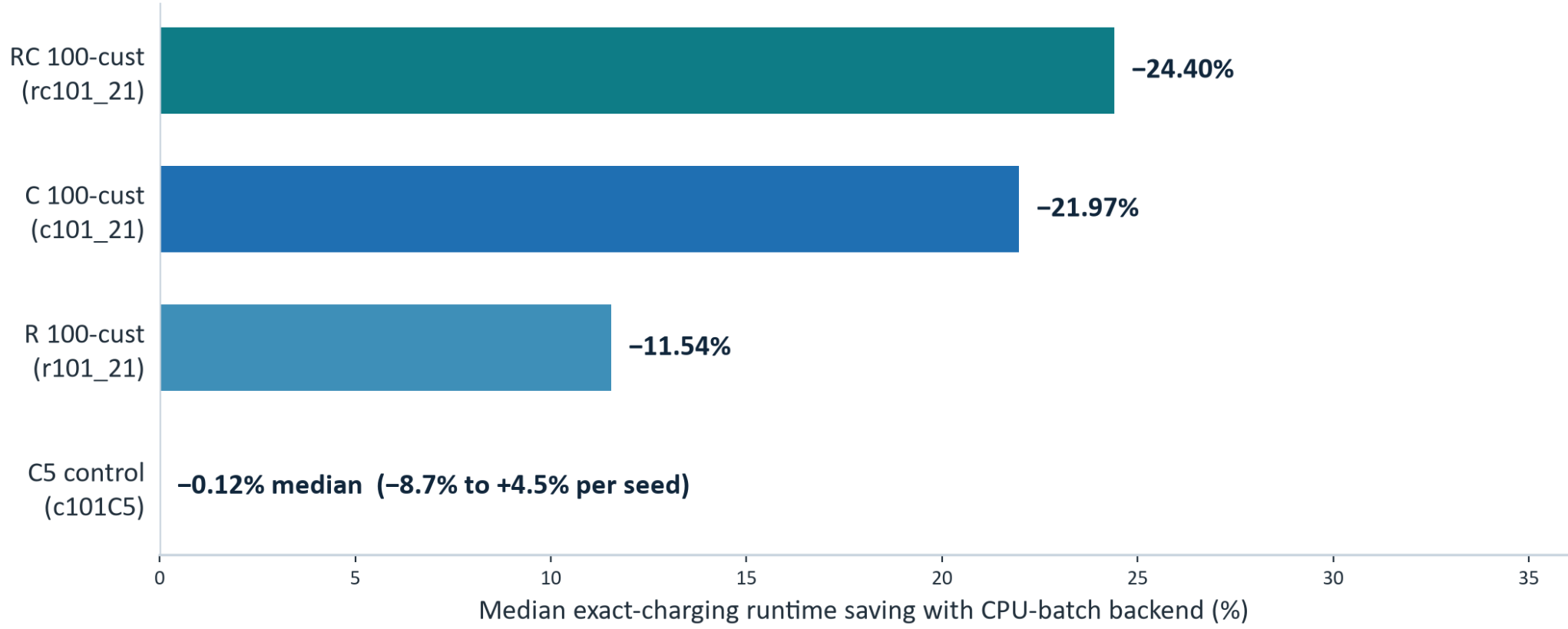
READY

Results — Fleet Quality (12 instances × 3 seeds)



Vehicle-first search cuts fleet size on 5 of 12 instances while distance improves

Results — Exact-Charging Runtime



CPU-batch ships as default after independent audit; control instance shows noise-level change

Why It Works

- Vehicle-count-aware repair — route elimination first, insertion into existing routes; fleet size can fall but never rise
- Shared optimistic prefilters (capacity · time · energy) screen before exact evaluation — only changed routes re-solved
- Constraint-guided destroy ranks station pressure, time-window conflicts, worst energy detours — removal lands where constraints bind
- Stage 4 adaptive operator weights beat fixed weights on 7/36 runs — kept as an observable lane, not the default
- Lexicographic objective — vehicles first: every acceptance step rejects any candidate that increases fleet size
- Independent review CLI replays raw solutions, event logs & hashes before any number is published

Honest Failure Analysis

- 92 of 124 published BKS entries are model-incompatible (fixed-fleet variants) — we report no gaps against them
- C5 control: –0.12% median (spread –8.7% to +4.5%) — speedup is structure-dependent, not universal
- Adaptive weights win on 7/36 runs only — the shipped default remains fixed weights

What we don't claim: no unattested baselines, no improvement beyond published evidence. Every number is replayable from raw JSON + event logs via the review CLI.

References

- [1] Schneider, Stenger & Goeke (2014) Transp. Sci. 48(4):500–520
- [2] Pisinger & Ropke (2006) Transp. Res. B
- [3] Keskin & Çatay (2016) Transp. Res. E
- [4] Hottung, Kwon & Tierney (2022) EJOR

Take-aways

- A full reproduction of SSG (2014) anchors every claim — Stage 0 is frozen and hash-verified
- Quality and runtime both improved under the stage-gated pipeline: –12.6% vehicles · –5.1% distance · –22% CPU
- All evidence — successes and failures — is published and independently re-playable

How to verify: every number in this poster replays from raw JSON + event logs via the independent review CLI — scan the QR (top right) for code & evidence.